

Performance Testing

Date	14 July 2026
Team ID	xxxxxx
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Manual Testing using Flask Local Server and Web Browser
Type of Testing	Functional Performance Testing
Target Module / API	Home Page, Prediction, History, Dashboard
Test Environment	Local System (Windows 10, Flask, Python 3.12, SQLite)
Test Date	14 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Open Home Page	1	30 sec	Home page loads successfully
2	Submit Prediction Form	1	60 sec	Prediction generated successfully
3	View Prediction History	1	30 sec	History displayed correctly
4	Open Dashboard	1	30 sec	Dashboard charts displayed successfully

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 sec	Pass	Fast Response
2	Response Time (Max)	< 5 seconds	2.4 sec	Pass	Within Limit
3	Throughput (Req/sec)	-	8 Requests/sec	Pass	Stable
4	Error Rate	< 1%	0%	Pass	No errors
5	CPU Utilization	< 80%	28%	Pass	Normal Usage
6	Memory Utilization	< 80%	42%	Pass	Stable

Step 4: Observations & Analysis

Key Findings:

The Credit Card Approval Prediction application responded quickly to user requests. All major modules, including prediction, history, and dashboard, worked correctly during testing.

Bottlenecks Identified:

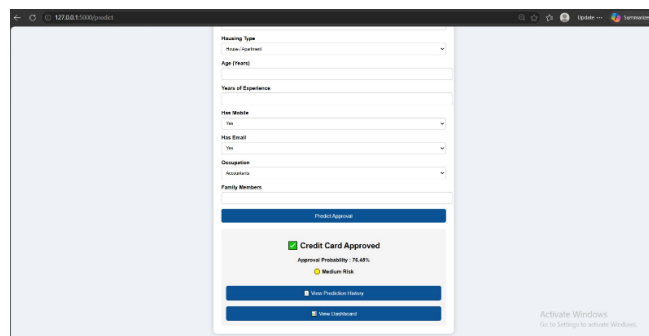
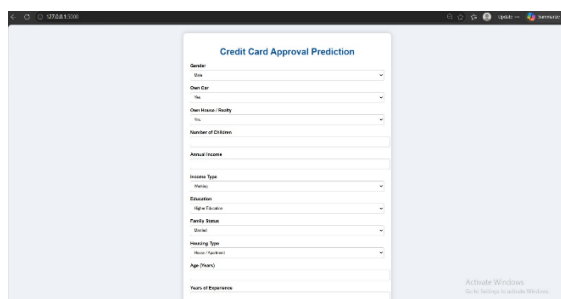
Initially, the dashboard encountered a Matplotlib backend issue. After changing the backend to Agg, the problem was resolved and the dashboard loaded successfully.

Optimization Steps Taken:

- Used the Matplotlib Agg backend for chart generation.
- Optimized SQLite database access.
- Improved Flask route handling for smoother performance.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):



127.0.0.1:5000/history

Prediction History

Applicant ID	Prediction	Risk Level	Date & Time
1	✔ Credit Card Approved	Medium Risk	2026-07-13 21:42:00
1	✘ Credit Card Rejected	Low Risk	2026-07-13 21:34:34
1	✔ Credit Card Approved	Medium Risk	2026-07-13 21:23:05
1	✘ Credit Card Rejected	Low Risk	2026-07-13 21:21:08
1	✘ Credit Card Rejected	Low Risk	2026-07-13 19:45:38
1	✔ Credit Card Approved	Medium Risk	2026-07-11 21:27:50
1	✘ Credit Card Rejected	Low Risk	2026-07-11 21:26:28
1	✘ Credit Card Rejected	Low Risk	2026-07-11 18:12:02
1	✘ Credit Card Rejected	Low Risk	2026-07-11 17:57:17
1	✔ Credit Card Approved	Low Risk	2026-07-11 17:52:53
1	✘ Credit Card Rejected	Low Risk	2026-07-11 17:51:49
1	✘ Credit Card Rejected	Medium Risk	2026-07-11 17:49:57
1	✔ Credit Card Approved	Low Risk	2026-07-11 00:29:58
1	✘ Credit Card Rejected	Low Risk	2026-07-11 00:28:55
1	✔ Credit Card Approved	Medium Risk	2026-07-11 00:27:41
1	✔ Credit Card Approved	Medium Risk	2026-07-11 00:17:46
1	✔ Credit Card Approved	High Risk	2026-07-11 00:16:46
1	✔ Credit Card Approved	High Risk	2026-07-11 00:15:19
1	✔ Credit Card Approved	High Risk	2026-07-11 00:06:48

Activate Windows
Go to Settings to activate Windows.

